

Detection of Fake Online Recruitment Using Machine Learning Techniques

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Abstract—There are a numerous amount of job postings on the internet and sometime these vacancy postings turns out to be fake. Even on the reputed and trusted job advertising platforms, people fall prey to these fake advertisements. After selection in the job, hiring people start demanding for money and details of bank account. Good number of candidates gets duped and lose loads of money and sometimes even their current jobs. So it would be very helpful to identify if the job listed on the website is real or fake. In this paper, we have used machine learning to detect fraud job vacancy postings.

In our proposed ML technique, we have applied two data balancing techniques namely “Adaptive Sympathetic” and “Synthetic Minority Oversampling Technique” in combination with “Term Frequency-Inverse Document Frequency” which is a feature extraction method. However, in the literature, some research have used “Bag of Words” for extraction of features which could just count the no. of times of the word appeared whereas the technique used in this research work (i.e TF-IDF) also provides the importance of words. The public “Employment Scam Aegean Dataset” (EMSCAD) was used which contained around 18000 job listings out of which about 800 are fraud listings. We have used two machine-learning models such as Random Forest and k- nearest classifiers to detect the online job is face or real. We have compared the performance of proposed models with existing models and it performs better in terms of recall, precision and f1-scores.

Keywords—Adaptive Sympathetic, Synthetic Minority Oversampling, Term Frequency-Inverse Document Frequency, Feature Extraction, EMSCAD.

I. INTRODUCTION

With all the improvements and development in technology, systems have now become more online to ease accessibility to the user. But this also increases chances of people getting tricked as they have very less knowledge about these new technologies. Online frauds are becoming quite common these days. People lack knowledge regarding protection of their systems. One category of online frauds is Fake posting of jobs on the internet. Naive people who are desperate to get jobs usually get duped by scammers. People will get e-mail saying that they have been selected for job but then scammers will ask for payment of money to start the process of joining. People who are innocent will pay hefty amounts of money thinking they will get the job but after payment they stop getting any reply from scammers posing as recruiter. This results a loss of individual's money and time. To stop this trickery we have built a detector system which would find out if the job is fake or legitimate with the help of various machine learning techniques.

In recent years, some companies have begun to publish job openings on the internet so that job seekers can access them as soon as feasible. Employment fraud is one of the most serious issues that has arisen in recent years. Fake job

postings will also harm the platform's reputation. This could, however, be a hoax by fraudsters who supply work to job seekers in exchange for money. While social networking platforms are an innovation in the computer sector, but then again, these platforms may be manipulated to fake information with ease, such as listing fake employment opportunities. There is proof that false information spreads more quickly and widely through social media channels because individuals can publish it multiple times with no verification of the information. To begin, supervised machine learning algorithms are being investigated as categorizing methods to address the problem of spotting scammers on job advertising

When people connect with misleading content on the internet, they are persuaded because they believe it is true. Most people does not even try to double-check the facts. To prevent any more fraud cases and loss of huge amount of money to an individual and defamation of big corporations, we definitely need to come out with solutions. Hence, we decided to detect these online bogus job postings using ML algorithms like “K-nearest neighbour (KNN)” and “Random Forest”. Different famous job recruitment platforms where corporation posts for job vacancy are Linked in, Glassdoor, Monster, indeed, naukri.com and many more. A fake job detector for this platform would be beneficial for individuals looking for the employment. While doing analysis of dataset, we found out the words (Fig 1) that are mostly used in fraudulent online job listings. Bigger the size of word, more frequently is was used in description of Job.



Fig 1: Frequently used words in Fake Job Description.

II. BACKGROUND STUDY

On the IT job dataset, Van et al [1] employed multiple deep neural network models such as "TextCNN, Bi-GRU-LSTM-CNN, and Bi-GRU-CNN" with usually multi word embeddings to investigate job prediction. They also proposed an easy and powerful ensemble model that combines many DNN models. They utilise the dataset to anticipate Papachristo's suggested computer jobs. This dataset contains 10,000 separate job descriptions tagged with 25 different sorts of computer-related jobs that were accessed from internet finding job websites. They developed a job prediction tool to assist IT students in finding a job that matches their knowledge, skills, interests, and other factors.

The research work [2], proposes to employ several approaches on mining of data and classifying algorithms, such as "KNN, decision tree (DT), support vector machine (SVM), naive bayes classifier(NB), random forest classifier, multilayer perceptron(MP), and deep neural network(DNN)", to determine whether a employment posting is either genuine or fake. They employed the 18000-sample 'Employment Scam Aegean Dataset' (EMSCAD). With three dense layers for deep neural classifier, deep neural network performs well for this classification challenge. The DNN model's data was trained using ten-fold cross validation, where 60% of the dataset was utilised for training, and 20% for assessing accuracy of models validation, and the left was used to evaluate the model's performance. The DNN model's mean categorizing accuracy (folds=10) is equal to 97.7%.

Habiba et al [3] recommended using ANN to detect online recruitment fraud. The proposed model is first trained and then tested using the public "Employment Scam Aegean Dataset" (EMSCAD) and adequate text preparation techniques. The activation function of ANN is ReLU, and there are 32, 16, and 8 neurons in each of the three main hidden layers. To prevent overfitting, we inserted a 0.2 probability dropout layer after each hidden layer. Finally, because this is a zero and one classification (Binary) task, the output layer is added with one neuron using the sigmoid activation function. The percentage of correctly anticipated fraud incidents out of all actual fraud instances is 96.02 percent.

The research work in [4], have done the case study, that address the challenge of detecting recruiting fraud and scams. The extreme gradient boosting method is used to build an effective recruiting fraud detection model. It creates an algorithm that extracts required information from job adverts and tests it on three different scenarios. It creates a sequence of steps for extracting required information from employment adverts, which is then tested on other different samples. It makes eight suggestions and then blends those suggestions with additional pool features. For feature extraction, it employs a two-step technique. Several models are used, with XGBoost having the highest accuracy of 97.94 percent.

For finding out the most significant words from the data file research work in [6] proposed a pair of feature extraction techniques, "CountVectoriser (CV)" and "term frequency-inverse document frequency (TFIDF)", were used individually. In order to recognize fake news, these variables were fed into "multinomial naive Bayes (MNB), random forest (RF), support vector classifier (SVC), and logistic regression (LR) classifiers", yielding eight classification

models. GridsearchCV was also used to find the testing and training scores for the parameters that were chosen.

Chakrabarty et al [7] manually collected 4000 data points, created a dataset, and then preprocessed the data. They employed useful features to forecast the outcome after filtering out irrelevant ones. They looked for specific factors for each job, such as accounting, financial sector, banking, sector, engineering, clothes, textiles, HR, organizational, production, hospital, health, IT, telecommunications, and many others, to ensure that their research was not confined to a single position. They next deleted stop words and used NLP called TF-IDF to process the resulting text data. On the processed text, machine learning classifier techniques were afterwards deployed.

III. PROPOSED METHODOLOGY

Before we begin implementation we must understand the all steps included in this research work. The data flow diagram of our research work in this paper is shown in the Fig. 2.

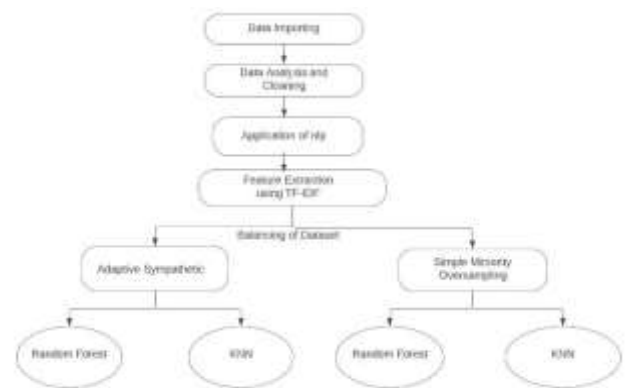


Fig 2: Flow diagram of proposed model.

Here we first imported data called "fake_job_postings.csv". Then we analysed and cleaned data. After cleaning process, we applied NLP process pipelines to convert the text understood by humans to a language that a computer machine could understand. Later, we used "TF-IDF" for extraction of important words with emphasis. There was a lot of imbalance between the two categories of data so we necessarily needed to balance the data before any application of model. We applied two models namely "KNN" and "Random Forest". Smote balancing technique with KNN seemed to perform best with 98 percent accuracy and descent f1 score. Next sections discuss each steps used in our research work in details.

A. Dataset

We used the benchmark dataset "false job listings" from the well-known website kaggle in this research work. The dataset we used is very skewed, since it contains 17,850 advertisement records, with roughly 17,000 authentic and around 860 fraudulent job postings. The corporation introduction, location, position explanation, job position heading, sector, perks, necessity qualification, kind of job, and roles were all employed in this study as independent variables for experiments. Table.1 show all the columns that were present in EMSCAD dataset when downloaded from Kaggle.

Table 1: Columns of EMSCAD Dataset

Job_id	Title
Department	Salary_range
Description	Requirements
Telecommuting	Has_company_logo
Location	Company_profile
Benefits	Has_questions
Employment_type	Required_experience
Required_education	Industry
Function	Fraudulent

B. Preprocessing of Data

Machine learning models were trained using text data in this research work. This text dataset has a lot of unclear content, which makes machine learning models difficult to use. This ambiguity can sometimes add to the complexity of the model's training, lowering its accuracy. Below are the steps that are followed in preprocessing of data..

a) Tokenization:

In this first step, paragraphs are broken into split sentences which further are separated into words. It is the procedure of splitting paragraph of text into a multiple small tokens. It can of three types: “word, character symbol, and sub-word”. This breakdown is necessary first step in NLP pipeline as this will reduce the complexity.

b) Lemmatization

Lemmatization means to providing actual root word with meaning using a “vocabulary and morphological” learning of words, with the aim of deleting only “inflectional endings”, providing the basic description of a word, called lemma. Stemming is also same as lemmatization but sometimes stemming does not provide us with root words, which are meaningless as shown in Fig. 3 below.



Fig 3: Distinction between Stemming and Lemmatization

c) Stop-Words/Punctuation/Numerical Elimination

Stopwords, such as he, it, do, us, the, and others, are parts of the text that help a human understand the statement in terms of appropriate syntax but don't contribute much information to the sentence. As a result, removing these stopwords can help to simplify the learning feature set. We utilised the NLTK library to create these stopwords. Numerical values and punctuations like commas etc., also be removed.

d) Conversion to Lower Case

Because all of the words have the same meaning, we transform them to lower case. Machine learning models, on the other side, are distinct from learning. For example, the words 'Dog' and 'dog' have the same context but they are distinct for machine learning models. As a result, we transform each term to small case in order to address the problem. This process was completed using the Python method “tolower()”.

C. Feature Extraction

We used an algorithm for extracting more important words that will improve accuracy of the models. “Bag of Words” (BOW) is a famous word extraction method but it does not provide us the knowledge of which word is more crucial. Another method for extracting characteristics which we have also used in this paper is the “term frequency-inverted document frequency” (TF-IDF). This method is frequently used to evaluate feature extraction methods during classification of text and its analysis. It assigns weight to tokens/words in a corpus depending on the document's inverse frequency and the incidence of the terms. Keywords with a higher weighted score are prioritised.

The influence of a word in a file is directly equal to the frequency of the words. Therefore,

$$TF(w, \text{sentence}) = \text{count of } w \text{ in sentence} / \text{number of words in sentence}$$

IDF means “inverse of document frequency”, which indicates how informative a “particular word” is. When we compute value for IDF, it is going to be less for most often appearing words, for example: stop words (since stop words like “is” appear all over practically in every text, and M/df will give that word a very small number).

$$IDF(\text{particular word}) = \log(M/(df + 1))$$

Finally, TF-IDF is a product of word frequency and document frequency which is inversed.

$$TF-IDF(w, \text{sentence}) = TF(w, \text{sentence}) * \log(M/(df + 1))$$

D. Balancing Of Data

Dataset was highly imbalanced so we used two different oversampling techniques namely ADASYN and SMOTE. Adaptive synthetic (ADASYN) seemed to perform same in case of accuracy but SMOTE seemed to perform a tad better than ADASYN in case of measurement metrics like recall and f1 scores.

i. Adaptive Synthetic Sampling Technique(ADASYN)

Firstly, The focus of “Adaptive Synthetic” is to create minority data samples based on their dispersal utilizing K-nearest neighbour. The algorithm adapts the dispersal without making any assumptions about the data's underlying distribution. For the KNN Algorithm, “Euclidean distance” is used. The big distinction between SMOTE and ADASYN is that the first one uses a density dispersal as a criteria to calculate the amount of new specimens that should be produced for each small size specimen by continuously updating the weightage of the low size samples to reimburse for bad distributions. On the other hand, the latter produces identical number of synthetic samples for minority sample. Table 2 shows number of fake and non fraudulent data points earlier and later on using ADASYN oversampling technique.

Table 2: Samples Quantity (ADASYN)

	Before	After
Fraudulent(1)	607	11954
Legitimate(0)	11909	11909

ii. Synthetic Minority Over sampling Technique (SMOTE)

Synthetic Minority method uses a K-nearest neighbour strategy, in which they finds K-nearest neighbour, combines them, and creates new synthetic samples so that binary data can be equal (balanced). The algorithm takes the word vectors and their closest neighbour and calculates their distance. The difference is used in a product with a certain value (which is random) between zero and one and then returned to the word. SMOTE is a game-changing algorithm from which many others have been derived. Table 3 shows samples initially and at end of SMOTE application.

Table 3: Samples Quantity (SMOTE)

	Before	After
Fraudulent(1)	627	11889
Legitimate(0)	11889	11889

E. Model Training, Testing and Validation

After we perform the data balancing, we fed that data into machine learning models like KNN and Random Forest. Data was split in train data(70%) and test data(30%). First ADASYN sampling was applied with KNN and RF

algorithms. Later on, above ML models were applied in sample data that was balanced with SMOTE technique. Used grid search cv in a way to get best parameters for model application. Parameters of KNN were `n_jobs=-1`, `n_neighbors=2`, `weights='distance'`. Parameters of Random Forest were `n_jobs=3`, `oob_score=True`, `n_estimators=100`, `criterion="entropy"`.

IV. EXPERIMENTAL RESULTS

Following are the results of different models with two different balancing techniques. Best combination among all models applied was SMOTE balancing method with K-nearest neighbour classification algorithm. Accuracy attained was 98% which is pretty good for classification purposes. Table 4 shows the results of past research. Most of the people have ignored the imbalanced dataset but in this research work, we take balancing as important step .

Table 4: Comparison with Past Research Accuracy)

Author	Feature Extraction Techniques	ML Algorithm	Accuracy(%)
Devsmi et al [10]	GloVe	Sequential Neural Network	97
Keerthana et al [5]	Categorical encoding	Multi layer perceptron	71
Habiba et al [2]	-	Deep Neural Network	97.7
Proposed work	TF-IDF	KNN	98

We applied two ML models on Data after using two balancing techniques: “ADASYN” and “SMOTE” and compared their overall results. Combination of different balancing technique with machine learning models are discussed below with their obtained results:

a) ADASYN+Random Forest

In Fig. 4 below shows the performance of random forest model with ADASYN balancing technique. Accuracy attained was 97% which is pretty good.

	precision	recall	f1-score	support
0	0.97	1.00	0.98	5105
1	0.97	0.38	0.54	259
accuracy			0.97	5364
macro avg	0.97	0.69	0.76	5364
weighted avg	0.97	0.97	0.96	5364

RF (ADASYN)

Fig 4: Results of Random Forest (ADASYN)

b) ADASYN+ K-Nearest Neighbour

KNN model performed a bit better than random forest when paired with adaptive sympathetic as shown in Fig. 5 below.

	precision	recall	f1-score	support
0	0.98	0.99	0.99	5104
1	0.79	0.67	0.72	260
accuracy			0.98	5364
macro avg	0.89	0.83	0.85	5364
weighted avg	0.97	0.98	0.97	5364

Execution time for KNN(ADASYN): 0.47542793353398644 min

Fig 5: Results of KNN (ADASYN)

c) SMOTE+ RandomForest

Random forest with SMOTE was better combination than with adaptive synthetic. Recall and F1 scores were more improved for this method (Fig. 6).

	precision	recall	f1-score	support
0	0.97	1.00	0.98	5089
1	1.00	0.42	0.59	275
accuracy			0.97	5364
macro avg	0.98	0.71	0.79	5364
weighted avg	0.97	0.97	0.96	5364

Execution time for RF: 0.34389146566390993 min

Fig 6: Results of Random Forest (SMOTE)

d) SMOTE+ K-Nearest Neighbour

This method was most successful one among all the permutations. We got high accuracy of 98 percent, and F1 scores of 76 was also pretty decent as shown in Fig. 7.

	precision	recall	f1-score	support
0	0.99	0.99	0.99	5125
1	0.79	0.73	0.76	239
accuracy			0.98	5364
macro avg	0.89	0.86	0.88	5364
weighted avg	0.98	0.98	0.98	5364

Execution time for KNN(SMOTE): 0.4750571767489115 min

Fig 7: Results of KNN (SMOTE)

V. CONCLUSION

The Detection of fraud online job postings was successfully done in this research work. Feature extraction techniques performed better in case of TF-IDF than Bag of Words (BOW). This TF-IDF was combined with different oversampling techniques like ADASYN and SMOTE. Further two models namely K-nearest Neighbour and Random Forest was applied to data. The combination of Smote with KNN performed better and as compared to the existing literature models with an accuracy of 98%. Also metrics like recall, F1 scores performed better in case of

smote. Our proposed models does not perform well for F1 scores and recall scores and their score are descent but not that great. Even though TF-IDF is far better than bag of words but it can't store the relationship among the words. Storage of relationship would definitely result in better results. We will apply some other "feature extraction" techniques in the near future and detect the Fake Online Recruitment with better accuracy, F1 score and Recall.

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